GetPhysical Master Asset: B2C App

Consolidated Specification, UX Matrix, & Database Flow • Version 3.0

Document Protocol: This document maps the absolute truth for the B2C React Native application. It combines the detailed UX prose format (Elements, Actions, States) with the corresponding Supabase database interactions (Matrices).

0. GLOBAL ARCHITECTURE & RULES

0.1 Navigation LOCKED

Fixed bottom bar, 5 tabs, visible on all main screens: (1) Sodia, (2) Discover, (3) Cards, (4) Track (Lift/Move/Food/Explore), (5) Profile. Users can change the default launch screen in Settings.

0.2 Roles on one account LOCKED

- **Individual:** default state for everyone.
- **Staff/Owner:** sees "Switch to Business Mode" in Settings (native QR scanner).
- **Coach:** sees "Switch to Coach Dash" in Settings (client roster).
- **Post As:** A toggle chooses which identity publishes content.

0.3 Security & Auth LOCKED

Two-token JWT architecture. Refresh token stored in Secure Enclave. High friction (biometrics) required only on "Confirm & Pay" and destructive account actions. Zero friction (no biometrics) for QR generation, which creates a dynamic hash expiring every 60s.

0.4 Verification Rules LOCKED

Tracker stamps are system-generated and uneditable. Challenge entries must come from logged tracker sessions. Lift challenges require video; Move challenges auto-sync GPS.

0.5 Actionable-object principle LOCKED

Facility tags, coach widgets, and challenges on posts always render as tappable CTAs (Pill buttons). They never navigate away; they open bottom sheets. CTAs are preserved on reposts.

0.6 Global Architecture Matrix

Component	UI Elements / Actions	System & Database Logic
Role Switching	"Switch to Business Mode/Coach"	Validates `public.user_roles` (array of `['individual', 'staff', 'coach']`).
Post As Toggle	Identity selector in post creator	Assigns `author_id` and `author_type` to the `posts` row.
Field Limits	Post: 280c. Bio: 150c. Entry: 150c.	Enforced via React Hook Form and Supabase Check constraints.

PHASE 1: IDENTITY, AUTH & ONBOARDING

UX Flows: Screens 1-5 LOCKED

- **Screen 1: Splash.** System check on launch. Token fresh: route to Sodia. Token stale: biometric prompt. No token: Auth Landing. NEW Force-update block and Offline cache state.
- **Screen 2: Auth Landing.** Elements: Phone number (+63), "Send OTP", "Continue with Apple/Google". States: invalid number, rate-limited. NEW Social-login edge case routes to phone collection.
- **Screen 3: OTP Verification.** Elements: 6-box numeric grid, auto-advance, paste support, "Resend in 30s". States: wrong code (shake), expired.
- **Screen 4: Basic Info.** Elements: Avatar, First/Last Name, Username. NEW DOB age gate. System validates username uniqueness live.
- **Screen 5: Preferences.** Elements: Home Location (search), Fitness Goals (multi-select pills). System writes to the Weighted Ranking Engine.
- **Screen 6: Find Friends.** Elements: "Sync Contacts", list of matches, "Follow", "Skip". NEW Permission primers precede all OS prompts (Location, Camera, Health).

1.1 Identity & Onboarding Matrix

Screen / Component	Elements & User Actions	States & Database Operations
Auth & Splash	No UI interactables.	Check Enclave for JWT. If Apple/Google SSO returns no phone, redirect to OTP for B2B deduplication.
OTP Verification	6-box grid. Timer to Resend.	RPC call to verify OTP. Success issues JWT. New users write to `public.profiles`.
Basic Info	Name, Username, DOB.	Username uniqueness checked via debounced RPC. Age Gate (13+ to join, under 18 blocked from payments).
Preferences	Location, Fitness Goals.	Writes to `user_preferences` JSONB. Seeds the cold-start weights for WRE.
Find Friends	"Sync Contacts", Follow buttons.	Hashes local contacts, runs array intersection against `profiles.phone_hash`. Writes to `follows`.

PHASE 2: SODIA (THE SOCIAL ENGINE)

2.1 Main Feed & Stories Engine LOCKED

- **Header Tray:** Horizontal row of avatars. Colored ring = unseen story. Grey = seen. User's avatar has a "+" to create.
- **Stories Create:** Full-screen camera. Tool tray: Text, Location Tag, Tracker Stamp. Location generates booking pill. NEW Post As toggle. Settings for audience/replies.
- **Stories Viewer:** Segmented progress bar. Floating pills open checkout sheet over story. Bottom input "Send Message". Own-story view shows viewer list and delete.
- **Standard Feed Post:** Avatar, Location Tag, Media. Dynamic CTA pill (e.g., "Drop In"). Action row (Like, Comment, Repost, Share).
- **Create Post:** Select media, caption, Tag Facility, Tag Friends, attach widgets. NEW Automated check-in draft triggers via push when VIP card is scanned at gym.
- **Comments & Repost:** Nested replies, @mentions. Repost sheet offers "Quick Repost" or "Quote". All tags/CTAs are preserved on reposts.

2.1 Sodia Architecture Matrix

Component	Elements & User Actions	States & Database Operations
Feed Assembly (WRE)	Infinite scroll. Base chronologic feed.	WRE SQL View injects a "Discovery Card" (Suggested Gym/Challenge) every 4th/5th slot based on tracker exhaust.
Dynamic CTA Pill	"Drop In" (Reg A) or "Book Court" (Reg B).	Pulls live pricing/status from `facilities` table. Tapping invokes Universal Checkout Sheet.
Stories Engine	Location Tags, Tracker Stamps (draggable overlays).	Coordinates/scale of stickers saved as JSON overlay array. Cron job sets `is_active = false` after 24h.
Repost & Quote	"Quick Repost" or "Quote". Embeds original.	**Preservation Rule:** All CTAs and stamps remain fully clickable on the reposted version.

WRE Algorithm Rule: The Weighted Ranking Engine uses deterministic SQL. Tracker logs silently update `user_weights`, multiplying the relevancy score of related Discovery Cards.

PHASE 2.B: CHALLENGES & MESSAGING

2.7 Verified PR Challenges LOCKED

- **Create Challenge:** Title, Type (Lift/Move), Metric (1RM/Pace), Date, Location Lock toggle, Eligibility.
- **Challenge Post:** Fiery "Accept Challenge" CTA. Shows participant count and mini-leaderboard.
- **Submit Entry:** Dynamic upload (Video for Lift, auto-sync for Move). Picker shows only verified recent logs. System overlays stamp and posts as linked reply.
- **Leaderboard:** Pinned in comments. Entries mirror to user Profile.

2.8 Activity & Inbox NEW

- **Activity (Heart):** Rows for likes, comments, requests, challenge alerts ("You were passed").
- **Inbox Main:** Search bar, Primary & Requests tabs. Swipe to mute/delete. FAB "New Message" supports multi-select for Group Chats.
- **Group Chats:** Header shows editable Group Name and stacked avatars. Participant labels above bubbles. Manage members via overflow.
- **Interactive Bubbles:** Shared stamps show "Congratulate". Shared to Coach shows "View Raw Data". B2B Handshake cards show "Review & Accept".

2.2 Challenges & Messages Matrix

Component	Elements & User Actions	States & Database Operations
Challenge Creation	Location Lock toggle, Type, Metric.	Writes to `challenges`. Location Lock forces the system to validate geofence or VIP check-in during submission.
Accept Flow	Tapping "Accept Challenge".	Injects a Bounty Card into the user's Track Tab (`user_bounties` table).
Group Chats	Multi-select friends. Edit Group Name.	Writes to `chat_rooms` (type: group) and `chat_participants`.
B2B Handshake	"[Facility] invited you to join as [Role]"	Updates `public.user_roles` and maps `user_id` to facility's staff registry.

PHASE 2.C & 3: PROFILES & DISCOVER

2.10 User & Facility Profiles LOCKED

- **User Profile:** Header with Active Roster badges (gym memberships). Metrics: Followers, Following, Streaks, Badges. Tabs: (1) Feed, (2) PRs/Entries with "View All Personal Bests", (3) Locker, (4) Albums, (5) Journey.
- **Journey Tab** NEW **:** Vertical timeline of Progress Pics, Journal Entries, Learn widgets. Private by default.
- **Facility Profile (Reg A):** Cover, Logo, Live Capacity ("12/50 active"). Primary: "Buy Day Pass". Tabs: Feed, Community, Coaches, Info, Challenges.
- **Coach Pro Profile:** Certs, Bio, Tags. Primary: "Train with Me". Tabs: Feed, Programs, Transformations.

3.1 Discover Mapbox Engine LOCKED

- **Map:** Full-screen Mapbox, branded pins. Filters (Gyms, Pickleball, Challenges, Coaches).
- **Pin Sheet:** Cover, distance, Challenge badge ("Accept Challenge"), Social proof ("3 friends checked in"). Primary: "View Facility".
- **Search/List:** Search overlay by category. List view sortable by Distance, Rating, Price. Location denied fallback (blurred map).

2.3 Profiles & Discover Matrix

Component	Elements & User Actions	States & Database Operations
User Badges & Streaks	Header metrics. Tap Badges for sheet.	Streaks calculated from consecutive days with `workout_logs`. PB button opens categorized list of all-time bests.
Journey Tab	Progress Pics, Text Journals.	Strictly private. Pulls from private Albums. Excluded from algorithmic recommendation surfaces.
Map Pins	Standard (Logo), Event (Fire), Coach (Temp).	Queries `facilities` coordinates. Dynamic pins render based on active `challenges` at that location.
Location Fallback	Blurred map. Prompt for Zip Code.	Applies `blurView` to map canvas. Hard blocks discovery until coordinate seed is established.

PHASE 4: CARDS (VIP WALLET & SOA)

4.1 Cards Hub & Action Row NEW

- **3D Wallet Stack:** Vertical carousel of active passes. Tap flips 180° to show ticking 60s hash QR. Screen brightness auto-maxes.
- **Action Row:** Beneath the center card: [+] Add, [➡] Send, [₱] Pay.
- **Send Flow:** Tap Send -> select transferable pass -> select friend -> confirm transfer.
- **Pay Flow:** Tap Pay -> native scanner opens to scan merchant QR -> deducts from linked payment method.
- **Fitness SOA:** Bottom-sheet ledger scrolling over cards. Unified list (Payments, Expiries, Check-ins, Waitlists). Tap row for receipt detail.

4.2 Join Facility & Universal Checkout LOCKED

- **Join Modal:** "Scan QR" (standee) or Search Directory.
- **Checkout Sheet:** Inventory list (Day Pass, Sub, Court). Court extras (Time, Date, Players). Summary line with Service Fee. "Confirm & Pay" requires biometric.
- **Success State:** Injects pass into stack, prompts "Share to Sodia".

4.3 Cards & Wallet Matrix

Component	Elements & User Actions	States & Database Operations
QR Generation	Tap card to flip. Ticking countdown.	QR is a dynamic hash of `user_id` + `timestamp` expiring every 60s. Zero friction.
Send Action	Select pass, select friend, confirm.	Reassigns `owner_id` in `passes` table. Writes "Gift Sent/Received" to both users' ledgers.
Pay Action (POS)	Scan merchant cafe/merch QR.	Triggers POS payment flow. Deducts from linked method, writes transaction to SOA ledger.
Universal Checkout	Radio list. "Confirm & Pay" (Biometric).	Handles Xendit xenPlatform split logic. OPEN (APP E): Fixed ₱15 for one-time passes, flat 5% on subs.

PHASE 5: TRACK (TRINITY ENGINE - LIFT & EXPLORE)

5.6 Explore Tab (Library) NEW

- **Elements:** Workout of the Day (WOD) card, Featured Plans carousel, Exercise Bank grid (by muscle group), Featured Coaches row.
- **Actions:** Tapping a plan opens a summary sheet with brief text instructions and GIF form references. Heavy follow-along videos are restricted to encourage logging over passive watching.

5.2 Lift Tracker LOCKED

- **Dash:** Weekly checkmarks, Active Program card, "Quick Start", "Templates". Active Bounties inject as "Log Attempt" cards.
- **Active Workout:** Exercise block, set/rep/weight grid. Set types (warm-up, drop, failure). Checkmark turns green, starts floating Rest Timer. Auto-saves draft.
- **Complete Modal:** Confetti UI, Metrics row, PR flags, "Share to Sodia" button (generates Lift Stamp).
- **Tools:** 1RM calculator, Exercise DB search, Periodization programmer (4/8/12 weeks).

5.1 Lift & Explore Matrix

Component	Elements & User Actions	States & Database Operations
Explore Content	WOD Card, Featured Plans.	Closes "blank canvas" gap. Video playback restricted to lightweight looping GIFs for form reference.
Bounty Injection	"Log Attempt" cards on Lift Dash.	Accepted challenges read from `user_bounties`. Starting from bounty tags the sets to the challenge.
Active Workout	Grid, Set types, Rest Timer.	Auto-saves draft every set. Validates `personal_bests` table on completion to flag PRs.
Periodization	Program Setup (Linear or %-Based).	Auto-calculates loads for future weeks based on Epley 1RM math and saves to `programs`.

PHASE 5: TRACK (TRINITY ENGINE - MOVE & FOOD)

5.3 Move Tracker LOCKED

- **Dash:** Weekly mileage ring, "Start GPS", "Sync Device".
- **Universal Selector:** Grouped grid (Endurance, Water, Racket, Multisport, NEW Padel).
- **Live GPS:** Mapbox trail, Distance, Pace. Background tracking enforced.
- **Live Multisport/Court:** Swaps metrics by leg. Court mode shows pulsing timer + HR ring. Post-match prompts Win/Loss.
- **Complete Modal:** Map route, Splits table. NEW **Heart-Rate Zones UI:** Stacked bar chart (Z1-Z5).

5.1 Food Tracker LOCKED

- **Dash:** 4 Macro widgets, Hydration counter, Quick-Log slider ("My Meals"). Tap Hydration to edit/undo.
- **Logging:** Nutritionix NLP search ("2 boiled eggs and rice"). Barcode scanner.
- **Target Calculator:** NASM formulas (BMR/TDEE). FAB "Share Daily Macros" generates closed-ring Stamp.

5.2 Move & Food Matrix

Component	Elements & User Actions	States & Database Operations
Live GPS	Mapbox trail, Pause/Lock actions.	Background location tracking. Crash recovery for unfinished runs.
Heart-Rate Zones	Stacked bar chart (Z1-Z5).	Derived from connected wearable (Apple Health/Garmin) time-series data logged during session.
Food Logging	Nutritionix Search, Barcode scanner.	NLP parsing maps to discrete food items. Saves offline cache for recent foods.
Target Calculator	Age, Sex, Weight, Height, Activity inputs.	Calculates TDEE. Writes to `user_targets`. Updates visual progress rings dynamically.

PHASE 6 & 7: SETTINGS AND SAFETY

6.1 Settings LOCKED

- **Business Ops:** Switch to Business Mode / Coach Dash. Identity chooser.
- **Account:** Payment Methods, Subscriptions (Pause/Cancel). Canceling confirms retained access.
- **Privacy:** Private Account toggle, Message/Tag limits, Default challenge visibility, WRE opt-out.
- **Data/App:** Theme, Launch Screen, Integrations (Apple Health/Garmin). Data Deletion enforces grace period.

7.1 Safety, Moderation & Global UX NEW

- **Report/Block/Mute:** Universally available via overflow menus. Blocks remove mutual visibility.
- **Challenge Integrity:** Auto-checks for tracker-origin, geofence, and extreme threshold jumps (auto-flags as "Under review").
- **Global States:** Loading skeletons, empty states, offline banners. Offline allows cached Sodia, log sets, track runs, view owned cards.

6.1 Settings & Safety Matrix

Protocol	Elements & User Actions	States & Database Operations
Subscriptions	Manage/Cancel recurring memberships.	Updates Xendit recurring schedules. UI confirms what user retains until period end.
Data Deletion	"Deactivate", "Delete Account".	Warns if active passes exist. High-friction FaceID required. Implements 30-day soft delete.
Report / Block	Reasons list (Spam, Fake Entry, Scam).	Writes to `reports`. Block writes to `blocked_users`, nullifying DB relations.
Anti-Cheat Logic	System validates thresholds.	**OPEN (APP E):** If a submitted lift jumps >15% from historical 1RM, auto-flags as "Under Review".

APPENDIX E: RAPID-FIRE SOLUTIONS FOR OPEN DECISIONS

These are the proposed logic solutions for the open items regarding business rules and platform integrity, to be finalized before development lock.

Item	Proposed Solution & Rationale
1. Fee Model	**Hybrid:** Fixed ₱15 for one-time payments (Day Passes) + flat 5% on recurring memberships. Prevents sticker shock on expensive monthly subs while securing margin on cheap passes.
2. Payment Provider	**Xendit.** xenPlatform API handles complex B2B split payments and sub-account routing cleaner than Maya.
3. Minors Policy	**13+ base.** Under 18 accounts are restricted to social, logging, and free clubs. Payments are hard-blocked to avoid liability.
4. Boosted Cards	**Post-launch (Phase 2).** Gyms will pay for visibility, but Day 1 implementation risks ruining WRE algorithm trust. Must label "Sponsored".
5. Coach Programs	**Free to Preview, Pay to Unlock.** Show structure and Week 1/Day 1 for free as a teaser to maximize conversion.
6. Duplicate Entries	**Replace best attempt.** Keeps leaderboards clean. If a user beats their own PR, it overwrites their existing row.
7. Court Challenges	**Delay to V1.1.** Launch focuses strictly on Lift and Move (GPS). Court validation requires separate logic too risky for Day 1.
8. Offline QR Fallback	**TOTP (Time-Based OTP).** Card flips to a 6-digit code on screen refreshing every 60s. Front desk types it into GymDash.
9. Close Friends	**Scrap for launch.** Sodia is a discovery engine. Users can toggle account to "Private" instead of dual-audience relations.
10. Repost Affiliates	**Track invisibly, build UI later.** Add `source_repost_id` to DB now, but delay building the payout UI until cash flow permits.
11. Anti-Cheat Threshold	**>15% jump from history.** If a submitted log jumps >15% over historical 1RM, flag as "Under Review" requiring admin approval.
12. Hiding Entries	**Hide allowed, deletion blocked.** System log remains immutable in DB, but an "eye" toggle hides runs from the public Trophy Case.